II. Factorise using the identity $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Example 3.27

Factorise: $8p^3 - 12p^2q + 6pq^2 - q^3$

Solution:

Given $8p^3 - 12p^2q + 6pq^2 - q^3$

This can be written as $(2p)^3 - 12p^2q + 6pq^2 - (q)^3$

Comparing this with $a^3 - 3a^2b + 3ab^2 - b^3 = (a-b)^3$ we get $a = 2p, b = q$

The given expression can be expressed as

$(2p)^3 - 3(2p)^2(q) + 3(2p)(q)^2 - (q)^3 = (2p - q)^3$

$= (2p - q), (2p - q), (2p - q)$ are the three factors.

Exercise 3.4

1. Factorise the following by taking out the common factor

(i) $18xy - 12yz$ (ii) $9x^5y^3 + 6x^3y^2 - 18x^2y$ (iii) $x(b-2c) + y(b-2c)$

(iv) $(ax + ay) + (bx + by)$ (v) $2x^2(4x-1) - 4x + 1$ (vi) $3y(x-2)^2 - 2(2-x)$

(vii) $6xy - 4y^2 + 12xy - 2yzx$ (viii) $a^3 - 3a^2 + a - 3$ (ix) $3y^3 - 48y$ (x) $ab^2 - bc^2 - ab + c^2$

2. Factorise the following expressions

(i) $x^2 + 14x + 49$ (ii) $y^2 - 10y + 25$ (iii) $c^2 - 4c - 12$ (iv) $m^2 + m - 72$ (v) $4x^2 - 8x + 3$

3. Factorise the following expressions using $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ identity

(i) $64x^3 + 144x^2 + 108x + 27$ (ii) $27p^3 + 54p^2q + 36pq^2 + 8q^3$

4. Factorise the following expressions using $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ identity

(i) $y^3 - 18y^2 + 108y - 216$ (ii) $8m^3 - 60m^2n + 150mn^2 - 125n^3$

Objective type Questions

5. Factors of $9x^2 + 6xy$ are

(A) $3y, (x+2)$ (B) $3x, (3x+3y)$ (C) $6x, (3x+2y)$ (D) $3x, (3x+2y)$

6. Factors of $4 - m^2$ are

(A) $(2+m)(2+m)$ (B) $(2-m)(2-m)$ (C) $(2+m)(2-m)$ (D) $(4+m)(4-m)$

7. $(x+4)$ and $(x-5)$ are the factors of _____

(A) $x^2 - x + 20$ (B) $x^2 - 9x - 20$ (C) $x^2 + x - 20$ (D) $x^2 - x - 20$

8. The factors of $x^2 - 5x + 6$ are $(x-2)(x-p)$ then the value of p is _____

(A) -3 (B) 3 (C) 2 (D) -2

9. The factors of $1 - m^3$

(A) $(1+m), (1+m+m^2)$ (B) $(1-m), (1-m-m^2)$

(C) $(1-m), (1+m+m^2)$ (D) $(1+m), (1-m+m^2)$

10. One factor of $x^3 + y^3$ is

(A) $(x - y)$ (B) $(x + y)$ (C) $(x + y)^3$ (D) $(x - y)^3$

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